

Probability - Set 1 - Solutions

Specification M7.1-M7.7.

Answer key

Question	Answer	Topic
1	D	Probability
2	C	Probability
3	F	Probability
4	A	Probability
5	C	Probability
6	F	Probability
7	D	Probability
8	A	Probability
9	E	Probability
10	A	Probability
11	D	Probability
12	A	Probability

Worked solutions

Question 1 PRO-01 · Probability · spec M7.7

A bag contains 4 red and 6 blue counters. Two counters are taken at random without replacement. Find the probability that they are different colours.

Answer: D

$$P(RB) + P(BR) = \frac{4}{10} \cdot \frac{6}{9} + \frac{6}{10} \cdot \frac{4}{9} = \frac{48}{90} = \frac{8}{15}.$$

Question 2 PRO-02 · Probability · spec M7.6

Two fair dice are thrown. Find the probability that the product of the two scores is odd.

Answer: C

The product is odd only if both scores are odd: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

Question 3 PRO-03 · Probability · spec M7.7

$P(A) = 0.6$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.3$. Which of the following statements is/are true?

- I A and B are independent.
- II A and B are mutually exclusive.
- III $P(A \text{ or } B) = 0.8$

Answer: F

$P(A)P(B) = 0.3 = P(A \text{ and } B)$, so independent (I). Not mutually exclusive since they can both happen (II false).
 $P(A \text{ or } B) = 0.6 + 0.5 - 0.3 = 0.8$ (III).

Question 4 PRO-05 · Probability · spec M7.2

A spinner has probability 0.15 of landing on red. It is spun 200 times. What is the expected number of times it lands on red?

Answer: A

$$0.15 \times 200 = 30.$$

Question 5 PRO-06 · Probability · spec M7.7

In a class of 30, 18 study French, 12 study German and 5 study both. A student is chosen at random. Given that the student studies French, what is the probability that they also study German?

Answer: C

Conditional on French (18 students), 5 also study German: $\frac{5}{18}$.

Question 6 PRO-07 · Probability · spec M7.6

Three fair coins are tossed. Find the probability of getting more heads than tails.

Answer: F

Need 2 or 3 heads: $\frac{3+1}{8} = \frac{1}{2}$ (by symmetry between heads and tails).

Question 7 PRO-09 · Probability · spec M7.4

The probabilities that a train is early, on time or late are 0.1, x and $2x$ respectively. Find the probability that the train is not late.

Answer: D

$$0.1 + 3x = 1 \Rightarrow x = 0.3; \text{ not late} = 0.1 + 0.3 = 0.4.$$

Question 8 PRO-10 · Probability · spec M7.7

A fair die is thrown repeatedly until a six appears. Find the probability that the first six appears on the third throw.

Answer: A

$$\left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} = \frac{25}{216}.$$

Question 9 PRO-11 · Probability · spec M7.6

Two cards are drawn at random from a standard pack of 52 without replacement. Which of the following expressions is the probability that both are aces?

Answer: E

Without replacement the second draw has 3 aces among 51 cards.

Question 10 PRO-12 · Probability · spec M7.7

Events A and B are mutually exclusive with $P(A) = 0.3$ and $P(B) = 0.4$. Find $P(\text{neither } A \text{ nor } B)$.

Answer: A

$$P(A \text{ or } B) = 0.7, \text{ so neither} = 0.3.$$

Question 11 PRO-04 · Probability · spec M7.7

A biased coin has probability p of landing heads. It is tossed twice. The probability of exactly one head is $\frac{3}{8}$. Find the possible values of p .

Answer: D

$$2p(1-p) = \frac{3}{8} \Rightarrow 16p^2 - 16p + 3 = 0 \Rightarrow (4p-1)(4p-3) = 0.$$

Question 12 PRO-08 · Probability · spec M7.7

A box contains n white balls and 3 black balls. Two balls are drawn without replacement. The probability that both are white is $\frac{5}{12}$. Find n .

Answer: A

$$\frac{n}{n+3} \cdot \frac{n-1}{n+2} = \frac{5}{12} \Rightarrow 12n^2 - 12n = 5n^2 + 25n + 30 \Rightarrow 7n^2 - 37n - 30 = 0 \Rightarrow (7n+5)(n-6) = 0, \text{ so } n = 6.$$